

Student Name:

Rohit Patel

Roll No:

12SCI002

Subject:

Physics

School / Institution:	Kendriya Vidyalaya No.1, Pune
Student Name:	Rohit Patel
Roll Number:	12SCI002
Class:	XII-B
Subject:	Physics
Barcode:	BAR102
Date of Examination:	15 March 2025
Session:	2024-25

I hereby declare that the answers written in this booklet are my own. I have not used any unfair means during the examination.

Candidate's Signature: _____

Invigilator's Signature: _____

Room No: _____

SECTION A — Multiple Choice / Very Short (1 mark each)

Q1. State Gauss's Law in electrostatics.

[1 mark]

Gauss's Law: The total electric flux through any closed surface is equal to $1/\epsilon_0$ times the net charge enclosed within that surface.

$$\Phi = Q_{\text{enc}} / \epsilon_0$$

This is one of Maxwell's four equations governing electromagnetism.

Marks	
Awarded	
/1	
Initials	

Q2. What is the SI unit of magnetic flux?

[1 mark]

SI unit of magnetic flux is Weber (Wb).

$$1 \text{ Wb} = 1 \text{ Tesla} \times 1 \text{ m}^2 = 1 \text{ V}\cdot\text{s}$$

Magnetic flux $\Phi = B \cdot A \cdot \cos\theta$ where θ is angle between B and area vector A.

Marks	
Awarded /1	
Initials	

Q3. Define the term 'work function' of a metal. [1 mark]

Work function (ϕ) is the minimum energy required to eject an electron from the surface of a metal.

It is a property of the metal surface. Measured in electron volts (eV).

For photoelectric effect: $h\nu = \phi + KE_{\text{max}}$

Marks	
Awarded /1	
Initials	

Q4. Name the phenomenon which shows the wave nature of light. [1 mark]

Diffraction and Interference of light show wave nature.

In Young's double slit experiment, interference pattern of bright and dark fringes is formed.

Diffraction occurs when light bends around corners of an obstacle.

Marks	
Awarded /1	
Initials	

SECTION B — Short Answer (2 marks each)

Q5. Derive an expression for the electric field due to an infinite plane sheet of charge. [2 marks]

Consider an infinite plane sheet with surface charge density σ .

Draw a Gaussian surface as a cylinder with flat faces parallel to sheet, each of area A.

By Gauss's law: $\Phi = Q/\epsilon_0$

Flux through curved surface = 0 (E perpendicular to area vector)

Student Name: Rohit Patel

Roll No: 12SCI002

Subject: Physics

Flux through two flat faces = $2EA$

$$\therefore 2EA = \sigma A/\epsilon$$

$$E = \sigma/(2\epsilon)$$

Direction: perpendicular to the sheet, away from positive charge.

Marks	
Awarded	
/2	
Initials	

Q6. State and explain Faraday's laws of electromagnetic induction.

[2 marks]

First Law: Whenever there is a change in magnetic flux linked with a conductor, an EMF is induced.

Second Law: The magnitude of induced EMF is directly proportional to the rate of change of magnetic flux.

$$e = -d\Phi/dt \text{ (negative sign indicates Lenz's law)}$$

Lenz's Law: The induced current opposes the change causing it (conservation of energy).

Marks	
Awarded	
/2	
Initials	

Q7. A proton and an alpha particle have the same kinetic energy. Compare their de Broglie wavelengths.

[2 marks]

$$KE = p^2/2m, \text{ so } p = \sqrt{2mKE}$$

$$\text{de Broglie wavelength } \lambda = h/p = h/\sqrt{2mKE}$$

For proton: $m_p = 1u$, charge = e

For alpha: $m_\alpha = 4u$, charge = $2e$

$$\lambda_p/\lambda_\alpha = \sqrt{(m_\alpha/m_p)} = \sqrt{4/1} = 2$$

$\therefore \lambda_p = 2\lambda_\alpha$. The proton has twice the wavelength of the alpha particle.

Marks	
Awarded	
/2	
Initials	

SECTION C — Long Answer I (3 marks each)

Q8. Draw a labelled diagram of a moving coil galvanometer and explain its working. [3 marks]

DIAGRAM DESCRIPTION (drawn on supplementary sheet):

- Rectangular coil ABCD wound on soft iron core
- Permanent horseshoe magnet with N and S poles
- Cylindrical soft iron core for radial field
- Hair springs for restoring torque
- Scale and pointer attached to coil

WORKING:

When current I flows through coil, torque $\tau = nBIA$ acts on it (n=turns, B=field, A=area).

Coil deflects until restoring torque of spring = deflecting torque.

$k\theta = nBIA$ where k = spring constant

$\theta = nBIA/k$ (deflection proportional to current)

The galvanometer gives direct reading of current through pointer.

Marks	
Awarded	
/3	
Initials	

SECTION D — Long Answer II (5 marks each)

Q9. Describe the construction and working of a transformer. Derive the expression for transformation ratio. [5 marks]

TRANSFORMER — CONSTRUCTION:

Student Name: Rohit Patel

Roll No: 12SCI002

Subject: Physics

3. CONTROL RODS: Made of Cadmium or Boron. Absorb excess neutrons to control reaction rate. Pushed in to slow down, pulled out to speed up.

4. COOLANT: Water or liquid sodium. Absorbs heat from fuel rods and transfers to steam generator. Prevents overheating.

5. **SHIELDING:** Thick concrete walls. Prevents harmful radiation from escaping. Protects workers and environment.

6. STEAM GENERATOR: Heat from coolant converts water to steam which drives turbines to generate electricity.

Marks	
Awarded	
/5	
Initials	

Student Name: Rohit Patel

Roll No: 12SCI002

Subject: Physics

*** END OF ANSWER BOOKLET ***

MARKS SUMMARY (For Examiner Use Only)

Section	Max Marks	Marks Obtained	Examiner Initials
Section A (1 mark × 4)	4		
Section B (2 marks × 3)	6		
Section C (4-5 marks)	varies		
Section D (5-6 marks)	varies		
TOTAL	100		